

Soumendu Majee

sites.google.com/view/soumendu-majee · linkedin.com/in/soumendu-majee
soumendu.majee.1@gmail.com · +1-765-838-9407

Research Interests

Computational Imaging · Efficient Deep Learning · Camera ISP · X-ray Computed Tomography

Education

Ph.D., Electrical and Computer Engineering *June 2021*
Purdue University, IN, USA

Advisors: Prof. Charles A. Bouman and Prof. Gregory T. Buzzard

Thesis: High Speed Imaging via Advanced Modeling

GPA: 4.0/4.0

B.Tech., Electronics and Electrical Communication Engineering *May 2014*
Indian Institute of Technology, Kharagpur, India
GPA: 9.1/10

Skills

Languages: Python, C++, C, C#, MATLAB, Cython, Bash

ML Frameworks: PyTorch, TensorFlow, Keras, JAX, SNPE

Tools: CUDA, OpenCV, L^AT_EX

Experience

Staff Researcher II *September 2025 – Present*

Staff Research Engineer *March 2024 – August 2025*

Senior Engineer, Research *April 2022 – February 2024*

MPI Lab, Samsung Research America, TX, USA

- Led the development and commercialization of Unified-Tone-Map across multiple Galaxy camera pipelines for Samsung Galaxy S25 and S26
- Led the development and commercialization of Under Display Camera pipeline for Samsung Galaxy Z Fold5 and Fold6
- Contributed to the development and commercialization of Expert-Raw for Samsung Galaxy S23 and S26
- Contributed to the development and commercialization of Tetra AI Zoom for Samsung Galaxy S24
- Contributed to the development and commercialization of Tetra High-resolution Photography for Samsung Galaxy S24
- Contributed to the development and commercialization of Under Display Camera for Samsung Galaxy Z Fold4
- Leading university research collaborations with Rice University

Postdoctoral Research Associate *July 2021 – April 2022*

Los Alamos National Laboratory, NM, USA

- Material Identification from Radiographs and Intrinsic Radiation
- Learning Noise-robust Features for Dynamic Reconstruction

Research Aide *July 2019 – September 2019*

Advanced Photon Source, Argonne National Laboratory, IL, USA

- *Coded Exposure for High Speed X-ray CT Imaging*: Developed CodEx, a synergistic combination of coded acquisition and reconstruction for high speed tomographic imaging

Graduate Research Assistant *January 2015 – June 2021*

Purdue University, IN, USA

- *Multi-Slice Fusion for 4D X-ray CT Reconstruction*: Developed a novel method that fuses multiple low-D Convolutional Neural Networks to implement a 4D image prior

- *Denoising Short-exposure Dynamic Radiographs*: Developed a deep-learning based transfer-learning method for denoising short-exposure dynamic radiographs
- *Multi-Domain Weighing for Metal Artifact Reduction in X-ray CT*: Developed a novel data and image domain weighing for reducing metal artifacts in X-ray CT
- *Multi-Orientation Fusion CT Reconstruction*: Developed a modular method to fuse information from multiple CT scans at different orientations to produce a joint reconstruction
- *Cone-beam X-ray CT Reconstruction of Additively Manufactured Parts*: Developed a high fidelity CT reconstruction method using scatter correction for defect detection in additively manufactured parts
- *Detection and Localization of Neurons in Fluorescence Microscopy*: Developed a novel neuron detection method to improve frame-rate of fluorescence microscopy imaging by selectively scanning neuron locations

Undergraduate Researcher

January 2014 – August 2014

Indian Institute of Technology, Kharagpur, India

- *Efficient Wideband Spectrum Sensing for Cognitive Radio*: Developed a novel spectrum estimation algorithm for spectrum sensing in cognitive radio

Research Intern

May 2013 – July 2013

University of Toronto, ON, Canada

- Detection of Human Thought from EEG (Electroencephalograph) Signals

Selected Publications

Zhang, M., Majee, S., Wang, C., Lee, S.J. and Sheikh, H., 2024. CoDISP: Exploring Compressed Domain Camera ISP with RGB-guided Encoder. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 5878–5888).

Yang, D., Kemp, C.A., Majee, S., Buzzard, G.T. and Bouman, C.A., 2024, October. Pixel-weighted multi-pose fusion for metal artifact reduction in x-ray computed tomography. In *2024 IEEE 26th International Workshop on Multimedia Signal Processing (MMSP)* (pp. 1–6). IEEE.

Balke, T., Davis Rivera, F., Garcia-Cardona, C., Majee, S., McCann, M.T., Pfister, L. and Wohlberg, B.E., 2022. Scientific Computational Imaging COde (SCICO). *Journal of Open Source Software*, 7(LA-UR-22-28555).

Majee, S., Aslan, S., Gursoy, D. and Bouman, C.A., 2022. CodEx: a modular framework for joint temporal de-blurring and tomographic reconstruction. *IEEE Transactions on Computational Imaging*, 8, pp. 666–678.

Majee, S., Balke, T., Kemp, C.A., Buzzard, G.T. and Bouman, C.A., 2021. Multi-slice fusion for sparse-view and limited-angle 4D CT reconstruction. *IEEE Transactions on Computational Imaging*, 7, pp. 448–462.

Balke, T., Majee, S., Buzzard, G.T., Poveromo, S., Howard, P., Groeber, M.A., McClure, J. and Bouman, C.A., 2018. Model-Based Cone-Beam Tomography with Scatter Correction. Manuscript in preparation for *IEEE Transactions on Computational Imaging*.

Majee, S., Balke, T., Kemp, C.A., Buzzard, G.T. and Bouman, C.A., 2019, May. 4D X-ray CT reconstruction using multi-slice fusion. In *2019 IEEE International Conference on Computational Photography (ICCP)* (pp. 1–8). IEEE.

Balke, T., Majee, S., Buzzard, G.T., Poveromo, S., Howard, P., Groeber, M.A., McClure, J. and Bouman, C.A., 2018. Separable models for cone-beam MBIR reconstruction. *Electronic Imaging*, 2018(15), pp. 181–1.

Majee, S., Ye, D.H., Buzzard, G.T. and Bouman, C.A., 2017. A model based neuron detection approach using sparse location priors. *Electronic Imaging*, 2017(17), pp. 10–17.

Majee, S., Ray, P. and Cheng, Q., 2015, November. Efficient wideband spectrum sensing using random projection. In *2015 49th Asilomar Conference on Signals, Systems and Computers* (pp. 141–145). IEEE.

Open Source Software

- **SVMBIR** — Super-Voxel Model Based Iterative Reconstruction. <https://github.com/cabouman/svmbir>
- **MBIRCONE** — Model-Based Iterative Reconstruction for cone-beam CT. <https://github.com/cabouman/mbircone>
- **SCICO** — Scientific Computational Imaging COde. <https://github.com/lanl/scico>

Invited Talks

Electronic Imaging	<i>January 2023</i>
Samsung Research America	<i>January 2022</i>
Oak Ridge National Laboratory	<i>March 2021</i>
Los Alamos National Laboratory	<i>February 2021</i>
Dept. of Energy Light-source Tomography Coordination Meeting	<i>February 2021</i>
Electronic Imaging	<i>January 2021</i>
Electronic Imaging	<i>January 2019</i>

Professional Volunteering

Paper Reviewer

- IEEE Transactions on Image Processing
- IEEE Transactions on Computational Imaging
- IEEE International Conference on Image Processing
- International Journal of Medical Physics Research and Practice
- Journal of Nondestructive Evaluation

Teaching Experience

Teaching Assistant

Fall 2014, Spring 2016, Spring 2017

Purdue University

- Linear Circuits Analysis (ECE 201)
- Digital Image Processing I (ECE 637)

Scholarships & Awards

- | | |
|---|-----------|
| • Received president's award for contributions in High-Mega-Pixel AI Multi-Frame Processing | 2024 |
| • Received CTO award for contributions in AI Multi-frame Zoom for Galaxy S24 Ultra | 2024 |
| • Awarded MITACS Globalink Research Fellowship | 2013 |
| • Awarded the Jagadis Bose National Science Talent Search Scholarship | 2010–2014 |
| • Ranked 1 in the WBJEE Engineering Entrance Examination | 2010 |

References

Available upon request